

Course Code	Course Title	L	T	P	C
BECE102L	Digital Systems Design	3	0	0	3
Pre-requisite	Nil	Syllabus version			
		1.0			
Course Objectives					
1. Provide an understanding of Boolean algebra and logic functions. 2. Develop the knowledge of combinational and sequential logic circuit design. 3. Design and model the data path circuits for digital systems. 4. Establish a strong understanding of programmable logic. 5. Enable the student to design and model the logic circuits using Verilog HDL.					
Course Outcome					
At the end of the course the student will be able to 1. Optimize the logic functions using and Boolean principles and K-map. 2. Model the Combinational and Sequential logic circuits using Verilog HDL. 3. Design the various combinational logic circuits and data path circuits. 4. Analyze and apply the design aspects of sequential logic circuits. 5. Analyze and apply the design aspects of Finite state machines. 6. Examine the basic architectures of programmable logic devices.					
Module:1	Digital Logic	8 hours			
Boolean Algebra: Basic definitions, Axiomatic definition of Boolean Algebra, Basic Theorems and Properties of Boolean Algebra, Boolean Functions, Canonical and Standard Forms, Simplification of Boolean functions. Gate-Level Minimization: The Map Method (K-map up to 4 variable), Product of Sums and Sum of Products Simplification, NAND and NOR Implementation. Logic Families: Digital Logic Gates, TTL and CMOS logic families.					
Module:2	Verilog HDL	5 hours			
Lexical Conventions, Ports and Modules, Operators, Dataflow Modelling, Gate Level Modelling, Behavioural Modeling, Test Bench.					
Module:3	Design of Combinational Logic Circuits	8 hours			
Design Procedure, Half Adder, Full Adder, Half Subtractor, Full Subtractor, Decoders, Encoders, Multiplexers, De-multiplexers, Parity generator and checker, Applications of Decoder, Multiplexer and De-multiplexer. Modeling of Combinational logic circuits using Verilog HDL.					
Module:4	Design of data path circuits	6 hours			
N-bit Parallel Adder/Subtractor, Carry Look Ahead Adder, Unsigned Array Multiplier, Booth Multiplier, 4-Bit Magnitude comparator. Modeling of data path circuits using Verilog HDL.					
Module:5	Design of Sequential Logic Circuits	8 hours			
Latches, Flip-Flops - SR, D, JK & T, Buffer Registers, Shift Registers - SISO, SIPO, PISO, PIPO, Design of synchronous sequential circuits: state table and state diagrams, Design of counters: Modulo-n, Johnson, Ring, Up/Down, Asynchronous counter. Modeling of sequential logic circuits using Verilog HDL.					
Module:6	Design of FSM	4 hours			
Finite state Machine(FSM):Mealy FSM and Moore FSM , Design Example : Sequence detection, Modeling of FSM using Verilog HDL.					
Module:7	Programmable Logic Devices	4 hours			
Types of Programmable Logic Devices: PLA, PAL, CPLD, FPGA Generic Architecture.					

Module:8		Contemporary issues	
		2 hours	